Bright Tuition Class

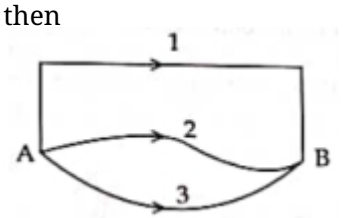
Subject : Physics
Standard : 11,12
Total Mark : 100

Jee physics full course 2

Paper Set : 1
Date : 11-01-2026
Time : 3H:0M

Physics - Section A (MCQ)

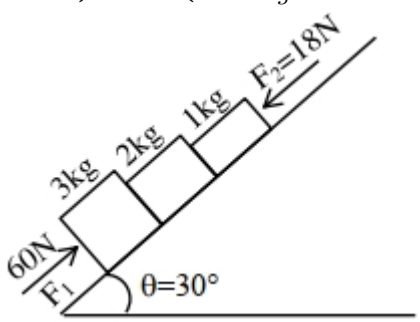
(1) A gravitational field is present in a region and a mass is shifted from A to B through different paths as shown. If W_1, W_2 and W_3 represent the work done by the gravitational force along the respective paths, then



(A) $W_1 = W_2 = W_3$ (B) $W_1 > W_2 > W_3$
(C) $W_1 > W_3 > W_2$ (D) $W_1 < W_2 < W_3$

(2) The angular speed of a fly wheel moving with uniform angular acceleration changes from $1200\,rpm$ to $3120\,rpm$ in 16 seconds. The angular acceleration in rad/s^2 is :
(A) 4π (B) 12π
(C) 104π (D) 2π

(3) In the diagram shown, the normal reaction force between $2\,kg$ and $1\,kg$ is (Consider the surface, to be smooth).....N (Given $g = 10\,ms^{-2}$)



(A) 25 (B) 39
(C) 6 (D) 10

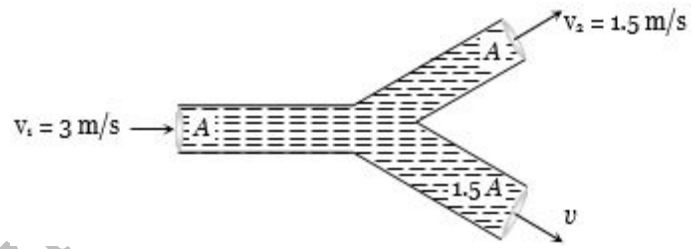
(4) The ratio of the distances travelled by a freely falling body in the $1^{st}, 2^{nd}, 3^{rd}$ and 4th second :
(A) $1:4:9:16$ (B) $1:3:5:7$
(C) $1:1:1:1$ (D) $1:2:3:4$

(5) The dimensions $[MLT^{-2}A^{-2}]$ belong to the :
(A) self inductance (B) magnetic permeability
(C) electric permittivity (D) magnetic flux

(6) If $\vec{F} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k}$, then the scalar and vector products of $\vec{F}$ and $\vec{r}$ have the magnitudes respectively as
(A) $5, \sqrt{3}$ (B) $4, \sqrt{5}$
(C) $10, \sqrt{2}$ (D) $10, 2$

(7) A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively :
(A) $\frac{S}{4}, \frac{3gS}{2}$ (B) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
(C) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$ (D) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

(8) An incompressible liquid flows through a horizontal tube as shown in the following fig. Then the velocity v of the fluid is .....m/s

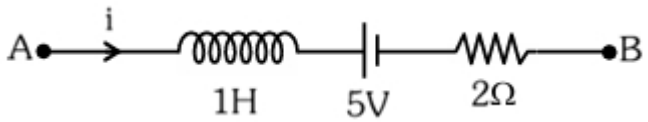


(A) 3 (B) 1.5
(C) 1 (D) 2.25

(9) To an ac power supply of $220\,V$ at $50\,Hz$, a resistor of $20\,\Omega$, a capacitor of reactance $25\,\Omega$ and an inductor of reactance $45\,\Omega$ are connected is series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively
(A) $7.8\,A$ and $30°$ (B) $7.8\,A$ and $45°$
(C) 15.6 and $30°$ (D) 15.6 and $45°$

(10) De−Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to (Given Bohr radius $= 0.052$ nm)
(A) $0.067\,nm$ (B) $0.67\,nm$
(C) $1.67\,nm$ (D) $2.67\,nm$

(11) AB is a part of an electrical circuit (see figure). The potential difference ' $V_A - V_B$ ', at the instant when current i $= 2$ A and is increasing at a rate of 1amp/ second is :



(A) 5 volt (B) 6 volt
(C) 9 volt (D) 10 volt

(12) A 2 amp current is flowing through two different small circular copper coils having radii ratio $1:2$. The ratio of their respective magnetic moments will be
(A) $1:4$ (B) $1:2$
(C) $2:1$ (D) $4:1$

(13) A microscope has an objective of focal length $2\,cm$, eyepiece of focal length $4\,cm$ and the tube length of

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$40\ cm$. If the distance of distinct vision of eye is $25\ cm$, the magnification in the microscope is :

(A) 100 (B) 125

(C) 150 (D) 250

(14) An electric dipole with dipole moment $5 \times 10^{-6}\ Cm$ is aligned with the direction of a uniform electric field of magnitude $4 \times 10^5 N/C$ The dipole is then rotated through an angle of $60°$ with respect to the electric field. The change in the potential energy of the dipole is

(A) $0.8\ J$ (B) $1.0\ J$

(C) $1.2\ J$ (D) $1.5\ J$

(15) An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then

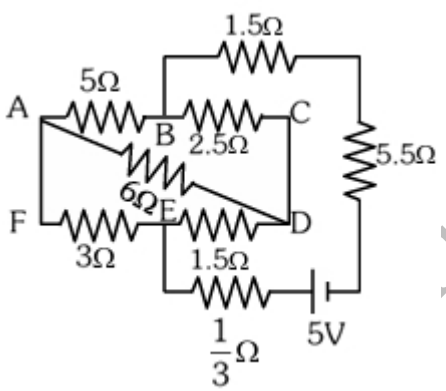
(A) reflected light is completely polarized and the angle of reflection is close to $60°$

(B) reflected light is partially polarized and the angle of reflection is close to $30°$

(C) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to $60°$ and $30°$, respectively

(D) transmitted light is completely polarized with angle of refraction close to $30°$

(16) The current passing through the battery in the given circuit, is :



(A) $2.0\ A$ (B) $0.5\ A$

(C) $2.5\ A$ (D) $1.5\ A$

(17) The distance between the two plates of a parallel plate capacitor is doubled and the area of each plate is halved. If C is its initial capacitance, its final capacitance is equal to

(A) $2C$ (B) $C/2$

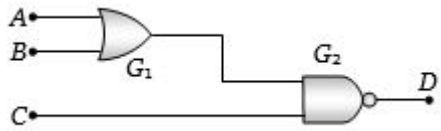
(C) $4C$ (D) $C/4$

(18) The mass of proton is $1.0073\ u$ and that of neutron is $1.0087\ u$($u =$ atomic mass unit). The binding energy of ${}^4_2He$ is (Given : helium nucleus mass $\approx 4.0015\ u$)

(A) $28.4\ MeV$ (B) $0.0305\ J$

(C) $0.0305\ erg$ (D) $0.061\ u$

(19) For the given combination of gates, if the logic states of inputs A, B, C are as follows $A = B = C = 0$ and $A = B = 1, C = 0$ then the logic states of output D are



(A) $0, 0$ (B) $0, 1$

(C) $1, 0$ (D) $1, 1$

(20) A galvanometer of $100\,\Omega$ resistance gives full scale deflection when $10\,mA$ of current is passed. To convert it into $10\,A$ range ammeter, the resistance of the shunt required will be ..... Ω

(A) -10 (B) 1

(C) 0.1 (D) 0.01

Physics - Section B (NUMERIC)

(21) Three masses are placed on the $x-$axis : $300\,g$ at origin, $500\,g$ at $x = 40\,cm$ and $400\,g$ at $x = 70\,cm$. The distance of the centre of mass from the origin is ..... cm

(22) A gas at NTP is suddenly compressed to one-fourth of its original volume. If γ is supposed to be $\frac{3}{2}$, then the final pressure is..... atmosphere

(23) Water fall from a $40\,m$ high dam at the rate of $9 \times 10^4\ kg$ per $hour$. Fifty percentage of gravitational potential energy can be converted into electrical energy. Using this hydroelectric energy number of $100\,W$ lamps, that can be lit, is
(Take $g = 10\,ms^{-2}$)

(24) The ratio of the power of a light source S_1 to that the light source S_2 is $2.S_1$ is emitting 2×10^{15} photons per second at 600 nm . If the wavelength of the source S_2 is 300 nm , then the number of photons per second emitted by S_2 is _____ $\times 10^{14}$.

(25) A double slit interference experiment performed with a light of wavelength $600\ nm$ forms an interference fringe pattern on a screen with 10th bright fringe having its centre at a distance of $10\ mm$ from the central maximum. Distance of the centre of the same 10th bright fringe from the central maximum when the source of light is replaced by another source of wavelength $660\ nm$ would be _____ mm .

Light Tuition Class

Bright Tuition Class

Subject : Chemistry
Standard : 11,12
Total Mark : 100

12th JEE Full Course -2

Paper Set : 1
Date : 11-01-2026
Time : 0H:0M

Chemistry - Section A (MCQ)

(26) Arrange the following elements in increasing order of first ionization enthalpy:

Li, Be, B, C, N

Choose the correct answer from the options given below:

(A) Li < B < Be < C < N (B) Li < Be < C < B < N

(C) Li < Be < N < B < C (D) Li < Be < B < C < N

(27) Assuming complete ionisation, same moles of which of the following compounds will require the least amount of acidified $KMnO_4$ for complete oxidation?

(A) $FeSO_3$ (B) FeC_2O_4

(C) $Fe(NO_2)_2$ (D) $FeSO_4$

(28) Match List I with List II.

List I (Compound)	List II (Shape/geometry)
A. NH_3	I. Trigonal Pyramidal
B. BrF_5	II. Square Planar
C. XeF_4	III. Octahedral
D. SF_6	IV. Square Pyramidal

Choose the correct answer from the options given below:

(A) $A-II, B-IV, C-III, D-I$

(B) $A-III, B-IV, C-I, D-II$

(C) $A-II, B-III, C-IV, D-I$

(D) $A-I, B-IV, C-II, D-III$

(29) The energy of an electron in the ground state $(n=1)$ for He^+ion is $-x$ J, then that for an electron in $n=2$ state for Be^{3+} ion in J is

(A) $-\frac{x}{9}$ (B) $-4x$

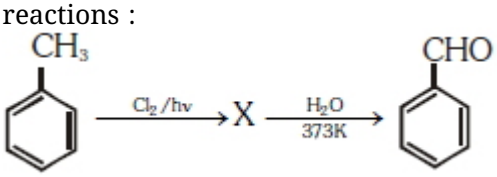
(C) $-\frac{4}{9}x$ (D) $-x$

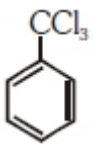
(30) The right option for the mass of CO_2 produced by heating $20g$ of 20% pure limestone is ..... g(Atomic mass of $Ca=40$)$\left[CaCO_3 \xrightarrow{1200K} CaO + CO_2\right]$

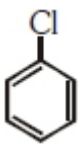
(A) 1.32 (B) 1.12

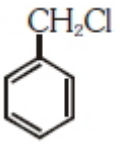
(C) 1.76 (D) 2.64

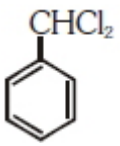
(31) Identify compound X in the following sequence of reactions :



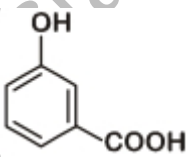
(A)


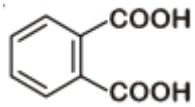
(B)


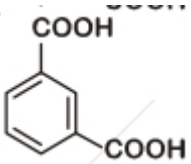
(C)


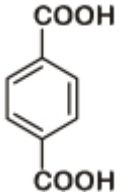
(D)


(32) Which of the following acid will form an (a) Anhydride on heating and (b) Acid imide on strong heating with ammonia?

(A)


(B)


(C)


(D)


(33) The compound A on treatment with Na gives B, and with PCl_5 gives C.B and C react together to give diethyl ether. A, B and C are in the order

(A) $C_2H_5OH, C_2H_6, C_2H_5Cl$

(B) $C_2H_5OH, C_2H_5Cl, C_2H_5ONa$

(C) $C_2H_5Cl, C_2H_6, C_2H_5OH$

(D) $C_2H_5OH, C_2H_5ONa, C_2H_5Cl$

(34) For a reaction, activation energy $E_a = 0$ and the rate constant at 200 K is 1.6×10^6 s^{-1}. The rate constant at 400 K will be glven that gas constant]
$R = 8.314$ $JK^{-1}mol^{-1}$

(A) 3.2×10^4 s^{-1} (B) 1.6×10^6 s^{-1}

(C) 1.6×10^3 s^{-1} (D) 3.2×10^6 s^{-1}

(35) In an S_{N^1} reaction on chiral centres, there is

(A) inversion more than retention leading to partial racemisation

(B) 100% retention

(C) 100% inversion

(D) 100% racemisation.

(36) Gadolinium has a low value of third ionisation enthalpy because of .....

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(A) high exchange enthalpy

(B) high electronegativity

(C) high basic character

(D) small size

(37) Which of the following reactions is appropriate for converting acetamide to methanamine?

(A) Hoffmann hypobromamide reaction

(B) Stephen's reaction

(C) Gabriel phthalimide synthesis

(D) Carbylamine reaction

(38) Statement I : Acid strength increases in the order given as HF $<$ HCl $<<$ HBr $<<$ HI

Statement II : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is correct but Statement II is false.

(D) Statement I is incorrect but Statement II is true.

(39) Identify the reaction from following having top position in EMF series (Std. red. potential) according to their electrode potential at $298\ K$.

(A) $K^+ + 1e^- \rightarrow K_{(s)}$ (B) $Mg^{2+} + 2e^- \rightarrow Mg_{(s)}$

(C) $Fe^{2+} + 2e^- \rightarrow Fe_{(s)}$ (D) $Au^{3+} + 3e^- \rightarrow Au_{(s)}$

(40) Strong reducing behaviour of H_3PO_2 is due to

(A) Low oxidation state of P

(B) Presence of one $-OH$ group and two $P-H$ bonds

(C) Presence of two $-OH$ groups and one $P-H$ bonds

(D) Low coordination number of P

(41) What is the correct electronic configuration of the central atom in $K_4\,[\mathrm{Fe(CN)_6}]$ based on crystal fleld theory?

(A) $t_{2g}^4 e_g^2$ (B) $t_{2g}^6 e_g^0$

(C) $e^3 t_2^3$ (D) $e^4 t_2^2$

(42) Which of the following reactions is the metal displacement reaction? Choose the right option.

(A) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

(B) $Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2Cr$

(C) $Fe + 2HCl \rightarrow FeCl_2 + H_2 \uparrow$

(D) $2\ Pb\,(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2 \uparrow$

(43) An organic compound contains $78\,\%$ (by $wt.$) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is : [Atomic wt. of C is 12, H is 1]

(A) CH (B) CH_2

(C) CH_3 (D) CH_4

(44) During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

(A) concentrated sulphuric acid (B) dilute nitric acid

(C) dilute sulphuric acid (D) dilute hydrochloric acid

(45) Which amongst the following is incorrect statement?

(A) C_2 molecule has four electrons in its two degenerate π molecular orbitals.

(B) H_2^+ ion has one electron.

(C) O_2^+ ion is diamagnetic.

(D) The bond orders of O_2^+, O_2, O_2^- and O_2^{2-} are $2.5, 2, 1.5$ and 1, respectively.

Chemistry - Section B (NUMERIC)

(46) In an experiment it showed that $10\ mL$ of $0.05\ M$ solution of chloride required $10\ mL$ of $0.1\ M$ solution of $AgNO_3$, which of the following will be the formula of the chloride (X stands for the symbol of the element other than chlorine)

(47) The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is

(48) The rate constant for a first order reaction is $4.606 \times 10^{-3} s^{-1}$. The time required to reduce $2.0\,g$ of the reactant to $0.2\,g$ is..... S

(49) Given that

$\Lambda_m^\alpha = 133.4\ (AgNO_3)\,; \Lambda_m^\alpha = 149.9(KCl)$

$\Lambda_m^\alpha = 144.9\,S\,cm^2\,mol^{-1}\ (KNO_3)$

the molar conductivity at infinite dilution for $AgCl$ is ..... $S\,cm^2\,mol^{-1}$

(50) The total number of orbitals present for principle quantum number, $n = 4$ is

Bright Tuition Class

BRIGHT TUITIONS CLASSES

Subject : Mathematics
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Full course 2

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Mathematics - Section A (MCQ)

(51) An organization awarded 48 medals in event 'A',25 in event 'B ' and 18 in event ' C '. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of three events?

(A) 10 (B) 9
(C) 21 (D) 15

(52) Let $f: R-\{2,6\} \rightarrow R$ be real valued function defined as $f(x)=\frac{x^2+2x+1}{x^2-8x+12}$. Then range of f is

(A) $\left(-\infty,-\frac{21}{4}\right] \cup[0, \infty)$ (B) $\left(-\infty,-\frac{21}{4}\right) \cup(0, \infty)$
(C) $\left(-\infty,-\frac{21}{4}\right] \cup\left[\frac{21}{4}, \infty\right)$ (D) $\left(-\infty,-\frac{21}{4}\right] \cup[1, \infty)$

(53) The number of real solutions of the equation $\mathrm{x}|\mathrm{x}+5|+2|\mathrm{x}+7|-2=0$ is .....

(A) 9 (B) 3
(C) 6 (D) 1

(54) Remainder when $64^{32^{32}}$ is divided by 9 is equal to .....

(A) 5 (B) 4
(C) 8 (D) 1

(55) If the first term of an $A.P.$ is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to

(A) -1200 (B) -1080
(C) -1020 (D) -120

(56) Let the lines $3x-4y-\alpha=0, 8x-11y-33=0$, and $2x-3y+\lambda=0$ be concurrent. If the image of the point $(1,2)$ in the line $2x-3y+\lambda=0$ is $\left(\frac{57}{13}, \frac{-40}{13}\right)$, then $|\alpha\lambda|$ is equal to :

(A) 84 (B) 101
(C) 113 (D) 91

(57) Let a circle C pass through the points $(4,2)$ and $(0,2)$, and its centre lie on $3x+2y+2=0$. Then the length of the chord, of the circle C, whose midpoint is $(1,2)$, is:

(A) $\sqrt{3}$ (B) $2\sqrt{3}$
(C) $4\sqrt{2}$ (D) $2\sqrt{2}$

(58) Three dice are rolled. If the probability of getting different numbers on the three dice is $\frac{p}{q}$, where p and q are co-prime, then $q-p$ is equal to

(A) 4 (B) 3
(C) 1 (D) 2

(59) Let the six numbers $a_1, a_2, a_3, a_4, a_5, a_6$ be in $A.P.$ and $a_1+a_3=10$. If the mean of these six numbers is $\frac{19}{2}$ and their variance is σ^2, then $8\sigma^2$ is equal to

(A) 220 (B) 210
(C) 200 (D) 105

(60) $\lim_{x \rightarrow \infty} \frac{\left(2x^2-3x+5\right)(3x-1)^{\frac{x}{2}}}{\left(3x^2+5x+4\right)\sqrt{(3x+2)^x}}$ is equal to:

(A) $\frac{2}{\sqrt{3e}}$ (B) $\frac{2e}{\sqrt{3}}$
(C) $\frac{2e}{3}$ (D) $\frac{2}{3\sqrt{e}}$

(61) If the system of equations $x+2y+3z=3$; $4x+3y-4z=4$; $8x+4y-\lambda z=9+\mu$ has infinitely many solutions, then the ordered pair (λ, μ) is equal to

(A) $\left(\frac{72}{5}, \frac{21}{5}\right)$ (B) $\left(\frac{-72}{5}, \frac{-21}{5}\right)$
(C) $\left(\frac{72}{5}, \frac{-21}{5}\right)$ (D) $\left(\frac{-72}{5}, \frac{21}{5}\right)$

(62) Let α and β be real numbers. Consider a 3×3 matrix A such that $A^2=3A+\alpha I$. If $A^4=21A+\beta I$, then

(A) $\alpha=1$ (B) $\alpha=4$
(C) $\beta=8$ (D) $\beta=-8$

(63) If $y=\frac{(\sqrt{x}+1)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}}+\frac{1}{15}\left(3\cos^2 x-5\right)\cos^3 x$, then $96y'\left(\frac{\pi}{6}\right)$ is equal to :

(A) 15 (B) 150
(C) 453 (D) 105

(64) If the function $f(x)=2x^3-9ax^2+12a^2x+1$, where a >0, attains its local maximum and local minimum values at p and q , respectively, such that $p^2=q$, then $f(3)$ is equal to:

(A) 55 (B) 10
(C) 23 (D) 37

(65) $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}} dx$ is equal to

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$
(C) $\frac{\pi}{6}$ (D) 2π

(66) Let $\int x^3 \sin x dx=g(x)+C$, where C is the constant of integration. If $8\left(g\left(\frac{\pi}{2}\right)+g'\left(\frac{\pi}{2}\right)\right)=\alpha\pi^3+\beta\pi^2+\gamma, \alpha, \beta, \gamma \in Z$, Then $\alpha+\beta-\gamma$ equals :

(A) 55 (B) 47
(C) 48 (D) 62

(67) Let $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}, \vec{b}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$. If the vector $\vec{c}$ is perpendicular to $\vec{b}$ and $\vec{a} \cdot \vec{c}=5$, then $|\vec{c}|$ is equal to

(A) $\frac{1}{3\sqrt{2}}$ (B) 18
(C) 16 (D) $\sqrt{\frac{11}{6}}$

(68) Let $P(3,2,3), Q(4,6,2)$ and $R(7,3,2)$ be the vertices of $\triangle$PQR. Then, the angle $\angle$QPR is

(A) $\frac{\pi}{6}$ (B) $\cos^{-1}\left(\frac{7}{18}\right)$

(C) $\cos^{-1}\left(\frac{1}{18}\right)$ (D) $\frac{\pi}{3}$

(69) Let (α, β, γ) be the foot of perpendicular from the point $(1, 2, 3)$ on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. then $19(\alpha + \beta + \gamma)$ is equal to :

(A) 102 (B) 101

(C) 99 (D) 100

(70) A company has two plants A and B to manufacture motorcycles. 60% motorcycles are manufactured at plant A and the remaining are manufactured at plant B. 80% of the motorcycles manufactured at plant A are rated of the standard quality, while 90% of the motorcycles manufactured at plant B are rated of the standard quality. A motorcycle picked up randomly from the total production is found to be of the standard quality. If p is the probability that it was manufactured at plant B, then 126p is

(A) 54 (B) 64

(C) 66 (D) 56

.... Mathematics - Section B (NUMERIC) ....

(71) If $\sin x + \sin^2 x = 1, x \in \left(0, \frac{\pi}{2}\right)$, then $\left(\cos^{12} x + \tan^{12} x\right) + 3\left(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x\right) + \left(\cos^6 x + \tan^6 x\right)$ is equal to

(72) The number of words, which can be formed using all the letters of the word 'DAUGHTER', so that all the vowels never come together, is

(73) Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3), x \in R$ Then $f'(10)$ is equal to .....

(74) Let $y = y(x)$ be the solution of the differential equation $2\cos x \frac{dy}{dx} = \sin 2x - 4y \sin x, x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right) = 0$, then $y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right)$ is equal to_____

(75) Let $\overrightarrow{a} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\overrightarrow{b} = 3\hat{i} - 5\hat{j} + \hat{k}$ and c be a vector such that $\vec{a} \times \vec{c} = \vec{c} \times \vec{b}$ and $(\overrightarrow{a} + \overrightarrow{c}) \cdot (\overrightarrow{b} + \overrightarrow{c}) = 168$. Then the maximum value of $|\vec{c}|^2$ is :